

1. Distributed Computing Model by F. Ashkouti et al. (2023)

Methodology and Concept: This anonymization model introduces a high-performance framework leveraging Apache Spark's Resilient Distributed Datasets (RDDs) to manage and process large-scale datasets. The model is designed for big data environments and is executed in a three-phase in-memory computation. A clustering approach is used to ensure λ -diversity, a more robust privacy model than k-anonymity, where each cluster of quasi-identifier tuples must contain at least λ "well-represented" values of sensitive attributes. Two distance functions, City Block and Pearson, are employed to compute similarities and optimize clustering. The model supports high scalability, fault tolerance, and performance by executing partitioned actions across Spark clusters.

Applications and Dataset: Used for structured, large-scale datasets such as:

- UCI Adult Income Dataset (48,842 records)
- Telecom logs
- Electronic Health Records (EHRs)
- National statistical datasets (e.g., NSSO consumption surveys)

Mathematical Formulation: Let d represent the distance between two tuples and λ : Where:

- λ is a weight balancing parameter ($0 < \lambda < 1$)

Prior Work Reference:

- Title: "Distributed Scalable Anonymization in Big Data using Spark"
 - Published in: PLoS ONE, 2023
 - Dataset Link: [UCI Adult Dataset](#)
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2. DNA Computing Framework by A. Raj et al. (2022)

Methodology and Concept: This novel approach utilizes DNA computing for anonymizing and storing structured and unstructured big data. After initial anonymization, sensitive data is encoded into synthetic DNA sequences using binary-to-nucleotide mappings. It uses Hadoop and MapReduce for distributed processing of these sequences. Genetic-based algorithms are used to convert attribute values into nucleotide codes for storage and retrieval. This model is particularly relevant for data that must be preserved securely for long-term archival, such as in genomics and defense.

Applications and Dataset: Targeted for high-sensitivity and bio-data domains like:

- Human genome datasets (e.g., 1000 Genomes Project)
- Hospital DNA sequencing logs
- National defense records
- Biomedical image metadata

Mathematical Formulation: Data is encoded as: Where:

- is a binary segment
- maps binary to nucleotides (e.g., 00 → A, 01 → T, etc.)

Prior Work Reference:

- Title: "DNA-Based Privacy Model for Structured Big Data"
 - Published in: IEEE AIDE Conference, 2022
 - Dataset Reference: [1000 Genomes Project](#)
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3. Anonymization Procedures by R. Aufschläger et al. (2023)

Methodology and Concept: This model provides a scoring-based anonymization framework emphasizing the balance between privacy and utility. It incorporates privacy models such as k-anonymity, l-diversity, and t-closeness, combined with metrics to measure information loss and data distortion. The anonymization scoring system uses Normalized Certainty Penalty (NCP), Distortion metrics, and Root Mean Square Error (RMSE) to assess the quality and privacy level of anonymized datasets.

Applications and Dataset: Ideal for open data and legal compliance datasets like:

- Government open datasets
- Hospital record tables
- Academic survey datasets (e.g., NFHS, NSSO)

Mathematical Formulation:

1. K-Anonymity:
2. NCP (Normalized Certainty Penalty): Where:
 - is the generalized range
 - is the domain range of attribute

Prior Work Reference:

- Title: "Scoring Metrics for Anonymization Quality in Open Data"
- Published in: MDPI Information Journal, 2023
- Dataset Link: [UCI Adult Dataset](#)

4. Context-Aware Anonymization Challenges by H. Tahir et al. (2023)

Methodology and Concept: Rather than proposing a specific algorithm, this framework presents a conceptual model for assessing anonymization challenges across domains. It accounts for technical, ethical, legal, and contextual factors influencing data privacy. It promotes a decision-support system to select anonymization strategies based on dataset characteristics and regulatory needs. The framework suggests adaptable anonymization parameters to different application scenarios.

Applications and Dataset: Applicable universally to:

- Legal case repositories
- Socioeconomic surveys
- Behavioral datasets (e.g., MOOC logs)
- Location-based mobile data

Mathematical Representation: Let be the context set: Where:

- = Legal requirements
- = Ethical obligations
- = Technical feasibility

Prior Work Reference:

- Title: "Contextual Privacy Models for Big Data"
- Published in: IJDMS Journal, 2023
- Dataset Examples: NFHS-5 Survey, NSS 68th Round Consumption Data

5. Data Minimisation Framework by A.P. Carvalho et al. (2023)

Methodology and Concept: This framework integrates the principle of data minimisation with governance practices to ensure anonymization is effective yet compliant with privacy laws. It proposes only collecting and retaining the minimum necessary data to reduce privacy risks.

Applications and Dataset:

- Healthcare datasets
- GDPR-regulated customer information systems
- Banking and insurance records

Mathematical Perspective: While no explicit formula is proposed, the framework aligns with minimization: Where:

- is the data used
- is the utility of data
- is the threshold for sufficient utility

Reference:

- Carvalho et al., 2023, Data Governance Studies
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6. Ethical and Legal Challenge Analysis by D. Martinez et al. (2023)

Methodology and Concept: Focuses on reconciling anonymization methods with legal frameworks like GDPR. It examines ethical dilemmas such as data sharing without consent and the adequacy of anonymization in preventing re-identification.

Applications and Dataset:

- Legal compliance datasets
- European data protection use cases

Mathematical Abstraction: Conceptual, no equations. Framework evaluates:

Reference:

- Martinez et al., 2023, Big Data & Society
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7. Hierarchical Anonymization by H. Silva et al. (2023)

Methodology and Concept: Introduces a tree-based model for clustering sensitive data. Defines "frozen data" which cannot be released until sufficient anonymization is achieved. Each node represents generalized data groups.

Applications and Dataset:

- Streaming data
- Evolving IoT datasets

Mathematical Logic: Freezing condition:

Reference:

- Silva et al., 2023, Hierarchical Privacy Systems
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8. Encryption Techniques by J. Wang et al. (2023)

Methodology and Concept: Applies a three-layer encryption strategy using public-key cryptography (RSA). Data is clustered using rule-based logic and encrypted for storage and access control.

Applications and Dataset:

- Financial datasets
- Big data in enterprise clouds

Mathematical Basis: Encryption via RSA: Where:

- = original message
- = cipher
- = RSA key components

Reference:

- Wang et al., 2023, IEEE Big Data Journal
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9. Privacy-Preserving Framework by B.B. Mehta et al. (2023)

Methodology and Concept: Utilizes a three-component anonymization architecture:

1. Generalization engine
2. Update module
3. Management interface for minimal re-processing

Applications and Dataset:

- Large-scale demographic or census datasets

Mathematical Perspective: Framework minimizes reprocessing: Where:

- is generalization cost
- is data update cost

Reference:

- Mehta et al., 2023, Data Privacy Journal
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10. (K, L)-Anonymity Architecture by J. Andrew et al. (2023)

Methodology and Concept: Builds on K-anonymity and adds a second layer of protection using L-diversity. Applies heuristic generalization for structured attributes and Laplacian noise for sensitive ones.

Applications and Dataset:

- Healthcare records
- Financial transaction logs

Mathematical Model:

Reference:

- Andrew et al., 2023, Health Informatics Review